

Nicolas Loizou

Assistant Professor, Johns Hopkins University

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Research Interests

Stochastic and Large-Scale Optimization, Min-Max Optimization and Variational Inequalities, Randomized Numerical Linear Algebra, Distributed and Federated Learning, Equilibrium Problems and Learning in Games, Mathematical Foundations of Machine Learning.

Academic Appointments

Assistant Professor, Johns Hopkins University

Jan. 2022 – present

Department of Applied Mathematics and Statistics & Mathematical Institute for Data Science

Secondary Appointments: Computer Science & Electrical and Computer Engineering

Institutes: MINDS, DSAI, ROSEI

IVADO Postdoctoral Research Fellow, Mila / Université de Montréal

Sept. 2019 – Dec. 2021

Quebec Artificial Intelligence Institute

Hosts: Dr. Simon Lacoste-Julien and Dr. Ioannis Mitliagkas

Additional Research Appointments

Visiting Scholar, Flatiron Institute, New York City

May 2026 – present

Visiting Researcher, Vector Institute, Toronto

May 2024 – May 2025

Research Intern, Meta Fundamental AI Research (FAIR), Montreal

Host: Michael Rabbat

Aug. 2018 – Dec. 2018

Education

PhD in Optimization and Operational Research, University of Edinburgh

2015 – 2019

School of Mathematics; *Advisor:* Dr. Peter Richtárik

MSc Computing, Imperial College London

2014 – 2015

BSc Mathematics, National and Kapodistrian University of Athens

2010 – 2014

Selected Awards & Honors

AFOSR Young Investigator Program (YIP) Award, Computational Mathematics

Project: Randomized and Scalable Iterative Algorithms for Extreme-Scale

2026

Computational Science: Design, Analysis, and Applications

NSF CAREER Award, CISE/CIF

Project: Structural and Algorithmic Foundations of Variational Inequalities in

2026

Machine Learning: From Principles to Scalability

Johns Hopkins Catalyst Award

2025

Competitive university-wide research award for early-career faculty.

CISCO Research Award

2023

Competitive industry research award (single PI)

Computational Optimization and Applications (COAP) Best Paper Prize

2020

Best paper published in the journal in 2020.

OR Society's Doctoral Award (Runner-up)

2019

Recognizing the most distinguished doctoral research in Operational Research in the UK.

Publications

According to Google Scholar, my work has received more than 4,000 citations and has an h-index of 25.

Coauthors marked with (p), (d), (m), and (i) were, respectively, my postdoctoral researchers, doctoral students, master's students, and interns at the time of the work.

Thesis

N. Loizou.

Randomized Iterative Methods for Linear Systems: Momentum, Inexactness, and Gossip.

Ph.D. Dissertation, The University of Edinburgh, 2019

OR Society's Doctoral Award (runner-up) for the "Most Distinguished Body of Research leading to the Award of a Doctorate in the field of Operational Research" in the United Kingdom.

Journal Papers

R. M. Gower, G. Garrigos, N. Loizou, K. Mishchenko, D. Oikonomou^(d), F. Schaipp

Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation
Transactions on Machine Learning Research, 2026

D. Nguyen, M. Imami, L. Zhao, J. Wu, A. Borhani, A. Mohseni, M. Khunte, Z. Zhong, V. Shi, S. Yao, Y. Wang, N. Loizou, A. C Silva, P. J Zhang, Z. Zhang, Z. Jiao, I. Kamel, W. Liao, H. Bai

Federated Learning for Renal Tumor Segmentation and Classification on Multi-Center MRI Dataset
Journal of Magnetic Resonance Imaging, 2025

S. Mukherjee, N. Loizou, S. U. Stich

Locally Adaptive Federated Learning

Transactions on Machine Learning Research, 2024

T. Zheng, N. Loizou, P. You, E. Mallada

Dissipative Gradient Descent Ascent Method: A Control Theory Inspired Algorithm for Min-max Optimization

IEEE Control Systems Letters, 2024

A. Khaled, O. Sebbouh, N. Loizou, R. M. Gower, P. Richtárik

Unified Analysis of Stochastic Gradient Methods for Composite Convex and Smooth Optimization

Journal of Optimization Theory and Applications (JOTA), 199(2), 499-540, 2023

R. D'Orazio, N. Loizou, I. Laradji, I. Mitliagkas

Stochastic Mirror Descent: Convergence Analysis and Adaptive Variants via the Mirror Stochastic Polyak Stepsize,

Transactions on Machine Learning Research, 2023

Z. Shi, N. Loizou, P. Richtárik, M. Takac

AI-SARAH: Adaptive and Implicit Stochastic Recursive Gradient Methods

Transactions on Machine Learning Research, 2023

N. Loizou, P. Richtárik.

Revisiting Randomized Gossip Algorithms: General Framework, Convergence Rates and Novel Block and Accelerated Protocols.

IEEE Transactions on Information Theory 67 (12), 8300 - 8324, 2021.

N. Loizou, P. Richtárik.

Convergence Analysis of Inexact Randomized Iterative Methods,
SIAM Journal on Scientific Computing 42 (6), A3979-A4016, 2020.

N. Loizou, P. Richtárik.

Momentum and Stochastic Momentum for Stochastic Gradient, Newton, Proximal Point and Subspace Descent Methods
Computational Optimization and Applications, 77 (3), 653-710, 2020. [COAP 2020 Best Paper Prize.](#)

Peer-reviewed Conference Papers

D. Oikonomou^(d), N. Loizou

Adaptive Sharpness-Aware Minimization with a Polyak-type Step size: A Theory-Grounded Scheduler
Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)

D. Oikonomou^(d), N. Loizou

Safeguarded Stochastic Polyak Step-Sizes for Nonsmooth Optimization: Robust Performance Under Small (Sub)Gradients
Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)

S. Choudhury^(d), N. Loizou

Extragradient Method for (L_0, L_1) -Lipschitz Root-Finding Problems
Advances in Neural Information Processing Systems 39 (NeurIPS 2025)

T. Yoon^(p), S. Choudhury^(d), N. Loizou

Multiplayer Federated Learning: Reaching Equilibrium with Less Communications
Advances in Neural Information Processing Systems 39 (NeurIPS 2025)

D. Oikonomou^(d), N. Loizou

Sharpness Aware Minimization: General Analysis and Improved Rates
12th International Conference on Learning Representations (ICLR 2025)

D. Oikonomou^(d), N. Loizou

Stochastic Polyak Step-sizes and Momentum: Convergence Guarantees and Practical Performance
12th International Conference on Learning Representations (ICLR 2025)

S. Choudhury^(d), N. Tupitsa, N. Loizou, S. Horvath, M. Takac, E. Gorbunov

Remove that Square Root: A New Efficient Scale-Invariant Version of AdaGrad
Advances in Neural Information Processing Systems 38 (NeurIPS 2024)

Y. Wang, P. Jian, Y. Dai, C. Jones, H. Sair, J. Shen, N. Loizou, J. Wu, W. Hsu, M. Imami, Z. Jiao, P. Zhang, H. Bai, **Enhancing vision-language models for medical imaging: bridging the 3D gap with innovative slice selection**
Advances in Neural Information Processing Systems 38 (NeurIPS 2024)

K. Emmanouilidis^(m), R. Vidal, N. Loizou

Stochastic Extragradient with Random Reshuffling: Improved Convergence for Variational Inequalities
27th International Conference on Artificial Intelligence and Statistics (AISTATS 2024)

S. Zhang^(p), S. Choudhury^(d), S. U. Stich, N. Loizou

Communication-Efficient Gradient Descent-Accent Methods for Distributed Variational Inequalities: Unified Analysis and Local Updates

11th International Conference on Learning Representations (ICLR 2024)

short version: Optimization for Machine Learning Workshop, NeurIPS 2022

S. Choudhury^(d), E. Gorbunov, N. Loizou

Single-Call Stochastic Extragradient Methods for Structured Non-monotone Variational Inequalities: Improved Analysis under Weaker Conditions

Advances in Neural Information Processing Systems 37 (NeurIPS 2023)

S. Sokota, R. D'Orazio, J.Z Kolter, N. Loizou, M. Lanctot, I. Mitliagkas, N. Brown, C. Kroer

A Unified Approach to Reinforcement Learning, Quantal Response Equilibria, and Two-Player Zero-Sum Games,

11th International Conference on Learning Representations (ICLR 2023)

short version: Deep Reinforcement Learning Workshop, NeurIPS 2022

A. Beznosikov⁽ⁱ⁾, E. Gorbunov⁽ⁱ⁾, H. Berard, N. Loizou

Stochastic Gradient Descent-Ascent: Unified Theory and New Efficient Methods,

26th International Conference on Artificial Intelligence and Statistics (AISTATS 2023),

short version: Optimization for Machine Learning Workshop, NeurIPS 2022

A. Orvieto⁽ⁱ⁾, S. Lacoste-Julien, N. Loizou

Dynamics of SGD with Stochastic Polyak Stepsizes: Truly Adaptive Variants and Convergence to Exact Solution,

Advances in Neural Information Processing Systems 36 (NeurIPS 2022)

Eduard Gorbunov⁽ⁱ⁾, Hugo Berard, Gauthier Gidel, Nicolas Loizou

Stochastic Extragradient: General Analysis and Improved Rates,

The 25th International Conference on Artificial Intelligence and Statistics (AISTATS 2022).

Eduard Gorbunov⁽ⁱ⁾, Nicolas Loizou, Gauthier Gidel

Extragradient Method: $O(1/K)$ Last-Iterate Convergence for Monotone Variational Inequalities and Connections With Cocoercivity,

The 25th International Conference on Artificial Intelligence and Statistics (AISTATS 2022).

Chris Junchi Li, Yaodong Yu, Nicolas Loizou, Gauthier Gidel, Yi Ma, Nicolas Le Roux, Michael I Jordan

On the Convergence of Stochastic Extragradient for Bilinear Games with Restarted Iteration Averaging,

The 25th International Conference on Artificial Intelligence and Statistics (AISTATS 2022),

short version: Optimization for Machine Learning Workshop, NeurIPS 2021 (**Oral Presentation**)

Nicolas Loizou, Hugo Berard, Gauthier Gidel, Ioannis Mitliagkas, Simon Lacoste-Julien

Stochastic Gradient Descent-Ascent and Consensus Optimization for Smooth Games: Convergence Analysis under Expected Co-coercivity

Advances in Neural Information Processing Systems 35 (NeurIPS 2021)

R. M. Gower, O. Sebbouh, N. Loizou

SGD for Structured Nonconvex Functions: Learning rates, Minibatching and Interpolation

The 24th International Conference on Artificial Intelligence and Statistics (AISTATS 2021),

short version: Optimization for Machine Learning Workshop, NeurIPS 2020

N. Loizou, S. Vaswani, I. Laradji, S. Lacoste-Julien

Stochastic Polyak Step-size for SGD: An Adaptive Learning Rate for Fast Convergence

The 24th International Conference on Artificial Intelligence and Statistics (AISTATS 2021)

short version: Optimization for Machine Learning Workshop, NeurIPS 2020 (**Spotlight Talk**)

N. Loizou, H. Berard, A. Jolicoeur-Martineau, P. Vincent, S. Lacoste-Julien, I. Mitliagkas

Stochastic Hamiltonian Gradient Methods for Smooth Games

Proceedings of the 37th International Conference on Machine Learning (ICML 2020)

A Koloskova, N Loizou, S Boreiri, M Jaggi, S. U. Stich

A Unified Theory of Decentralized SGD with Changing Topology and Local Updates

Proceedings of the 37th International Conference on Machine Learning (ICML 2020)

R. M. Gower, N. Loizou, X. Qian, A. Sailanbayev, E. Shulgin, P. Richtárik.

SGD: General Analysis and Improved Rates,

Proceedings of the 36th International Conference on Machine Learning (ICML 2019) (**Oral Presentation**)

M. Assran, N. Loizou, N. Ballas, M. Rabbat.

Stochastic Gradient Push for Distributed Deep Learning,

Proceedings of the 36th International Conference on Machine Learning (ICML 2019)

short version: Systems for ML workshop, NeurIPS 2018

N. Loizou, M. Rabbat, and P. Richtárik.

Provably accelerated randomized gossip algorithms.

International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 7505–7509, 2019.

N. Loizou, P. Richtárik.

Accelerated gossip via stochastic heavy ball method.

Annual Allerton Conference on Communication, Control, and Computing (Allerton), pages 927–934, 2018.

N. Loizou, P. Richtárik.

A New Perspective on Randomized Gossip Algorithms,

IEEE Global Conference on Signal and Information Processing (GlobalSIP), pages 440–444, 2016.

N. Loizou.

Distributionally Robust Games with Risk-Averse Players,

Proceedings of 5th International Conference on Operations Research and Enterprise Systems (ICORES), pages 186–196, 2016.

Preprints

T. Yoon^(p), N. Loizou

Per-Player Proximal Methods for Communication-Efficient Equilibrium Computation in Stochastic Multiplayer Games

PrePrint 2026

B. Balani^(d), T. Yoon^(p), N. Loizou

From Strategy Sharing to Increment Sharing: Per-Player Local Methods with Compression for Multi-player Federated Learning

PrePrint 2026

T. Yoon^(p), S. Choudhury^(d), E. Greenberg^(m), N. Loizou

Polyak-Type Extragradient Methods for Monotone Root-Finding Problems

PrePrint 2026

T. Yoon^(p), N. Loizou

Direct Acceleration of Stochastic Root-Finding Without Variance Reduction and Regularization

arXiv preprint arXiv:2608.12043

T. Yoon^(p), N. Loizou

On Same-Sample and Independent-Sample Stochastic Extragradient for Monotone Variational Inequalities

arXiv preprint arXiv:2608.06182

D. Oikonomou^(d), M. Buchholz, Y. Pun, R. M. Gower, N. Loizou

Taking the Road Less Scheduled with Adaptive Polyak Steps

arxiv preprint, arXiv:2511.07767

K. Emmanouilidis^(m), T. Ding, N. Nguyen, N. Loizou, R. Vidal

Certified Robustness from Approximate Gaussian Mixture Structures in Pretrained Latent Spaces

arXiv preprint arXiv:2605.25352)

Peer-reviewed Workshop Papers / Others

Siqi Zhang, Nicolas Loizou

ProxSkip for Stochastic Variational Inequalities: A Federated Learning Algorithm for Provable Communication Acceleration

Optimization for Machine Learning Workshop, NeurIPS 2022

R. D'Orazio, N. Loizou, I. H. Laradji, I. Mitliagkas

On Stochastic Mirror Descent: Convergence Analysis and Adaptive Variants

ICML 2021 - Workshop: Beyond first-order methods in ML systems

F. Hanzely, J. Konečný, N. Loizou, P. Richtárik, D. Grishchenko.

A Privacy Preserving Randomized Gossip Algorithm via Controlled Noise Insertion,

NeurIPS 2018 - Privacy Preserving Machine Learning Workshop

N. Loizou, P. Richtárik.

Linearly convergent stochastic heavy ball method for minimizing generalization error,

NeurIPS 2017 - Optimization for Machine Learning Workshop

F. Hanzely, J. Konečný, N. Loizou, P. Richtárik, D. Grishchenko

Privacy Preserving Randomized Gossip Algorithms

arXiv preprint arXiv:1706.07636, June 2017

N. Loizou.

Distributionally Robust Game Theory,

MSc Thesis, Imperial College London, 2015.

Mentoring and Advising

Current Members

- **Research Scientist**
 - TaeHo Yoon (Aug. 2026 - present)
- **Postdocs**
 - Ruhui Jin (Sept 2025 - present)
- **PhD Students:**
 - Bishnu Dev (Aug. 2026 - present)
 - Bhavika Balani (AMS) (Jan. 2026 - present)
 - Dimitrios Oikonomou (CS), (Sept. 2023 - present)
- **Undergraduate Students:**
 - Ezra Greenberg, (Sept 2025 - Present)

Past Members

- **Postdocs**
 - TaeHo Yoon (Dec. 2024 - Aug. 2026)
Next: Research Scientist, Optimization and Machine Learning Lab, Johns Hopkins University
 - Siqi Zhang (July 2022 - July 2024)
Next: Assistant Professor, Department of Industrial Engineering, Nanjing University.
- **PhD Student**
 - Sayantan Choudhury (AMS), (Sept. 2022 - Sept. 2025)
Next: Applied Scientist II, Amazon
- **MSE Students:**
 - Bhavika Balani, (Jan 2025 - Dec. 2025)
Next: PhD student, Department of Applied Mathematics and Statistics, JHU.
 - Jingyi Gu, (Jan 2025 - Dec. 2025)
 - Konstantinos Emmanouilidis (CS), (Sept. 2022 - Sept. 2024)
Next: PhD student, Department of Computer and Information Science, UPenn.
 - Mengtong Xu (June 2022 - Dec. 2022)
Next: PhD Student, Department of Computer Science and Technology, Tsinghua University.
 - Yichuan Wang (June 2022 - Dec. 2022)
Next: PhD Student, Department of Electrical and Computer Engineering, Boston University.
 - Zhichao Jia (June 2022 - Dec. 2022)
Next: PhD Student, School of Industrial and Systems Engineering, Georgia Tech.
- **Undergraduate Students:**
 - Sun Tanomkiattikun (Jan. 2025 - May 2025)
- **Research Interns:**
 - Eduard Gorbunov (June 2021 - Sept. 2021) from MIPT
 - Antonio Orvieto (June 2021 - Sept. 2021) from ETH Zurich
- **High-school Students:**
 - Christopher Davis (Summer 2024)
 - Carter Pisano (Summer 2023)

Grants and Research Support

Total research support to date: **\$1.90M** in awarded funding to my research, across projects totaling **\$2.39M**.

AFOSR Young Investigator Program (YIP) Award in Computational Mathematics — \$450,000, Sole PI

Randomized and Scalable Iterative Algorithms for Extreme-Scale Computational Science: Design, Analysis, and Applications 2026–2029

NSF CAREER Award — \$625,000, Sole PI

Structural and Algorithmic Foundations of Variational Inequalities in Machine Learning: From Principles to Scalability 2026–2031

JHU Catalyst Award — \$100,000, Sole PI

Reaching Equilibrium with Less Communication: Multi-Player Federated Learning for Multi-Agent LLM Systems 2025

NSF CISE Medium — \$900,000 total, PI; JHU share: \$525,000

Post-Modern Min-Max Optimization Theory: Departure from Classical Minimization Theory 2025–2028

JHU Discovery Award — \$100,000 total, MPI; Loizou share: \$25,000

Data Valuation in Distributed and Private Large-Scale Radiology 2024

Accelerated Translational Incubator Pilot Program — \$50,000 total, Co-PI; Loizou share: \$12,500

Enhancing Large Vision-Language Model Efficacy through Representative Slice Selection from 3D Medical Imaging Data 2024

Cisco Research Award — \$110,000, Sole PI

Multi-Player Federated Learning: Efficient Algorithms and Applications 2023–2025

JHU Bridge Grant — \$50,000, PI

2022

Earlier Fellowships & Scholarships

IVADO Postdoctoral Fellowship, Mila / Université de Montréal

Fully funded fellowship with **\$325,000 in total support**

(\$270,000 fellowship + \$45,000 research fund + \$10,000 additional support) Sept. 2019–Dec. 2021

Principal's Career Development Scholarship, University of Edinburgh

Highly competitive university-wide PhD scholarship 2015–2019

A.G. Leventis Foundation Scholarship

Scholarship supporting PhD studies at the University of Edinburgh 2015–2017

Imperial College London Department of Computing Scholarship

Merit-based scholarship awarded for MSc studies in Computing 2014–2015

Professional Service

Editorial Service

Information and Inference: A Journal of the IMA, Associate Editor, 2025–present

Optimization Methods and Software, Associate Editor, 2025–present

Transactions on Machine Learning Research, Action Editor, 2025–present

Area Chair Service

International Conference on Machine Learning (ICML): 2024, 2025, 2026

Annual Conference on Neural Information Processing Systems (NeurIPS): 2025, 2026

International Conference on Artificial Intelligence and Statistics (AISTATS): 2025, 2026

International Conference on Learning Representations (ICLR): 2027

Conference and Workshop Leadership

- **Cluster Chair**, “Continuous Optimization with Machine Learning,” International Symposium on Mathematical Programming (ISMP), Amsterdam, The Netherlands, 2027.
- **Mini-symposium Organizer**, SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom, June 2–5, 2026.
- **Organizing Committee**, DeepMath Conference, 2023 and 2024.
- **Mini-symposium Organizer**, “Adaptive, Line-Search, and Parameter-Free Stochastic Optimization,” International Symposium on Mathematical Programming (ISMP), Montreal, Canada, July 21–26, 2024.
- **Mini-symposium Organizer**, “Recent Advancements in Optimization Methods for Machine Learning,” SIAM Conference on Optimization, Seattle, WA, May 31–June 3, 2023.
- **Mini-symposium Organizer**, “Optimization Methods in Machine Learning,” 57th Annual Conference on Information Science and Systems (CISS), Baltimore, MD, March 22–24, 2023.
- **Cluster Co-Chair**, “Optimization for Data Science and Machine Learning,” International Conference on Continuous Optimization (ICCOPT), Lehigh University, Bethlehem, PA, 2022 (with S. Stich).
- **Mini-symposium Organizer**, “Recent Advancements in Optimization Methods for Machine Learning,” SIAM Conference on Optimization, Virtual Conference, July 20–23, 2021.
- **Session Chair**, “Stochastic Optimization Algorithms,” INFORMS Annual Meeting, Phoenix, AZ, November 2018.
- **Mini-symposium Organizer**, “Optimization Methods in Machine Learning,” 5th IMA Conference on Numerical Linear Algebra and Optimisation, University of Birmingham, United Kingdom, September 2016.
- **Mini-symposium Organizer**, “Randomized Numerical Linear Algebra,” 5th IMA Conference on Numerical Linear Algebra and Optimisation, University of Birmingham, United Kingdom, September 2016.
- **Seminar Organizer**, “All Hands Meetings on Big Data Optimization,” University of Edinburgh, 2015–2017.

External Reviewing

Grant Proposals:	<i>Department of Energy (DOE)</i> <i>National Science Foundation (NSF)</i>
Journals:	<i>Mathematical Programming,</i> <i>SIAM Journal on Optimization (SIOPT)</i> <i>Mathematics of Operations Research</i> <i>Journal of Machine Learning Research (JMLR)</i> <i>European Journal of Operational Research,</i> <i>Numerical Algorithms,</i> <i>Numerical Linear Algebra with Applications,</i> <i>IEEE Transactions on Signal and Information Processing over Networks,</i> <i>SIAM Journal on Mathematics of Data Science (SIMODS),</i>
Conferences:	<i>International Conference on Machine Learning (ICML): 2019-present</i> <i>Annual Conference on Neural Information Processing Systems (NeurIPS): 2020-present</i> <i>International Conference on Learning Representations (ICLR): 2019-present</i>
Workshops:	<i>Smooth Games Optimization and Machine Learning Workshop, NeurIPS 2019</i> <i>Optimization for Machine Learning Workshop, NeurIPS: 2020-present</i>

Johns Hopkins University Service

- **Chair**, AI-X Foundry Website Committee.
- **Chair**, MINDS Fellows Committee, 2023–present.
- **Organizer**, MINDS Weekly Seminar, 2023–present.

- AMS Communications & Marketing Committee, 2022–2024.
- MSE in Data Science Admissions Committee, 2022–2024.
- Data Science Postdoctoral Fellows Search Committee, 2022–2023.
- *Graduate Board Oral Examinations:*
 - Maryam Esmat (Physics & Astronomy), 2024
 - Wilson Gregory (AMS), 2024
 - Mao Hong (AMS), 2023
 - Ning Liu (AMS), 2023
 - Tianqi Zheng (ECE), 2023
- *PhD Thesis Committees:*
 - Mao Hong (AMS), 2024
 - Tianqi Zheng (ECE), 2024

Teaching

Courses marked with an asterisk () were designed and developed by me.*

Johns Hopkins University

- *Optimization for Data Science** (EN.553.662)
Spring 2026, Fall 2024, Fall 2023
- *Iterative Algorithms in Machine Learning: Theory and Applications** (EN.553.767)
Fall 2026, Fall 2024, Spring 2023
- *Large-Scale Optimization for Data Science** (EN.553.669)
Fall 2022

University of Edinburgh

- *Modern Optimization Methods for Big Data Problems*
Tutor: Spring 2018, Spring 2017
- *Optimization in Finance*
Tutor: Spring 2017
- *Fundamentals of Optimization; Simulation; Probability*
Tutor: Autumn 2016
- *Proofs and Problem Solving*
Tutor: Spring 2016

Invited Mini-Courses and Lecture Series

- *ELLIIT Symposium on Optimization for Learning*, Lund, Sweden
Mini-course: *Polyak-type Optimization: From Principles to Scalability** May 2026

Presentations

- Nov. 16–20, 2026, **SIAM Conference on Mathematics of Data Science (MDS26)**, Salt Lake City, Utah, U.S.
- 05 Nov, 2026 **Information Systems Laboratory (ISL) Colloquium**, Stanford University
- 01-04 Nov, 2026, **INFORMS Annual Meeting**, San Francisco, CA, USA
- 22 Oct 2026, **Mathematics of Machine Learning Seminar** at UMass Amherst
- 08 Oct, 2026, **FOLDS Seminar**, UPenn, USA
- 14 August, 2026, **AFOSR Computational Math Program Review**, Arlington, VA, USA
- 06-11 July, **ICML 2026**, Seoul, Korea
- 02–05 June, 2026: **Siam Conference on Optimization**, Edinburgh, UK.

- 01 June, 2026: **Oxford University, Machine Learning and Data Science Seminar**, Oxford, UK.
- 13 May, 2026: Mini-course Instructor **ELLIIT Symposium on Optimization for Learning**, Lund, Sweden.
- 06–08 May, 2026: **ELLIIT Symposium on Optimization for Learning**, Lund, Sweden.
- 17 Apr 2026, **Machine Learning Seminar**, Center for Computational Mathematics, Flatiron Institute, NYC.
- 27 Mar 2026: **LIONS seminar**, Arizona State University.
- 20-22 Mar 2026: **INFORMS Optimization Society (IOS) Conference**, Atlanta, Georgia, USA.
- 18 Feb 2026: **Physics Of Learning Seminar**, JHU, Department of Physics.
- 08 Dec 2025: **Applied Math Colloquium**, UCLA, Los Angeles.
- 02-07 Dec 2025: **NeurIPS**, San Diego, USA
- 24-28 April 2025: **ICLR 2025**, Singapore.
- 15 Jan 2025: **ROSEI Summit 2025**, Johns Hopkins University
- 25 Nov 2024: **Departmental Colloquium**, Rice CMOR, Rice University
- 21 Nov 2024: **Princeton Optimization Seminar**, Princeton University
- 12 Nov 2024: **CSE DSI Machine Learning Seminar Series**, University of Minnesota.
- 30 Oct 2024: **SILO Seminar**, University of Wisconsin, Madison
- 21-26 Jul 2024: **25th International Symposium on Mathematical Programming**, Montreal, Canada.
- 18 June 2024: **Vector Institute AI Seminar**, Vector Institute, Toronto, Canada.
- 08-11 May 2024: **ICLR 2024**, Vienna, Austria.
- 07 May 2024: **Computational Optimization seminar**, Imperial College London, UK
- 02-04 May 2024: **AISTATS 2024**, València, Spain.
- 22-24 Mar 2024: **INFORMS Optimization Society (IOS) Conference**, Rice University, Texas, USA.
- 10-16 Dec 2023: **NeurIPS 2022**, New Orleans, USA
- 20 Oct 2023: **Continuous Optimization Seminar**, University of Waterloo, Waterloo, Canada.
- 21 Sept 2023: **Flatiron Institute Seminar**, New York City, USA.
- 29 Aug 2023: **JHU ECE Seminar**, Baltimore, Maryland, USA.
- 10-14 Jul 2023: **14th International conference on Sampling Theory and Applications (SampTA 2023)**, Yale University, U.S.
- 31 May-03 Jun 2023: **SIAM Conference on Optimization (OP23)**, Seattle, Washington, U.S.
- 25-27 Apr 2023: **AISTATS 2023**, València, Spain
- 22-24 Mar 2023: **57th Annual Conference on Information Science and Systems**, Baltimore, USA
- 22-24 Feb 2023: **Coordinated Science Laboratory Student Conference (CSLSC)**, The University of Illinois at Urbana-Champaign, USA
- 28 Nov-03 Dec 2022: **NeurIPS 2022**, New Orleans, USA
- 25-28 Jul 2022: **International Conference on Continuous Optimization (ICCOPT) 2022**, Lehigh University, Bethlehem, Pennsylvania.
- 08 March 2022: **MINDS seminar**, Johns Hopkins University, Baltimore, US
- 06-14 Dec 2021: **NeurIPS 2021**, (virtual conference)
- 24-27 Oct 2021: **INFORMS annual meeting**, (virtual conference)
- 20-23 Jul 2021: **Siam Conference On Optimization**, (virtual conference)
- 07-09 Jul 2021: **18th International Workshop on Continuous Optimization**, (virtual conference)
- 28 April 2021: **Computer Science department**, Rice University.
- 13-15 Apr 2021: **24th International Conference on Artificial Intelligence and Statistics (AISTATS)**
- 23 Feb 2021: **University of British Columbia (Okanagan campus)**.
- 16 Feb 2021: **ECE Rising Star Seminar Series**, University of Wisconsin-Madison.
- 28 Jan 2021: **MINDS 2021 Winter Symposium**, Johns Hopkins University.
- 11 Dec 2020: **Optimization for Machine Learning Workshop**, NeurIPS, Spotlight Talk.
- 02 Dec 2020: **Montreal Machine Learning and Optimization (MTL MLOpt) Seminar**.

- 30 Nov 2020: **Smooth Games - Reading Group** EPFL, Virtual Talk.
- 07-13 Nov 2020: **INFORMS annual meeting**, Virtual Conference.
- 12-18 Jul 2020: **37th International Conference on Machine Learning (ICML)**, Virtual Conference
- 06-09 Jul 2020: **SIAM Conference on Imaging Science**, Virtual Conference
- 19 March 2020: **54th Annual Conference on Information Sciences and Systems (CISS)**, Virtual Conference.
- 06 Aug 2019: **International Conference on Continuous Optimization (ICCOPT)** , Berlin, Germany
- 11 Jun 2019: **36th International Conference on Machine Learning (ICML)** , Los Angeles, USA
- 14 May 2019: **IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)**, Brighton, United Kingdom
- 28 Apr 2019: **All Hands Meetings on Big Data Optimization** , KAUST, Saudi Arabia
- 8 Dec 2018: **Privacy Preserving Machine Learning Workshop, NIPS 2018** , Montreal , Canada
- 16 Nov 2018: **MILA Tea Talks** , University of Montreal , Montreal , Canada
- 13 Nov 2018: **Mathematics in Machine Learning Seminar** , McGill University, Montreal , Canada
- 04 Oct 2018: **56th Annual Allerton Conference on Communication, Control, and Computing (Allerton)** , Monticello, IL, USA
- 04 Sept 2018: **Facebook Montreal Research Seminar** , Montreal , Canada
- 13-17 Aug 2018: **DIMACS/TRIPODS/MOPTA** , Lehigh University, Bethlehem, PA, USA
- 1-6 Jul 2018: **International Symposium on Mathematical Programming** , Bordeaux, France (**invited**)
- 11 Apr 2018: **OptML group Seminar** , Lehigh University, Bethlehem, PA, USA
- 05 Feb 2018, **Optimization and Big Data 2018**, KAUST, Thuwal, Saudi Arabia,
- 8 Dec 2017, **NIPS Workshop on Optimization for Machine Learning**, Los Angeles, USA
- 05 Oct 2017, **RISELab, Department of Computer Science, UC Berkeley**, California, USA
- 31 May 2017, **Conference on Computational Management Science**, Bergamo, Italy, (**invited**)
- 22 May 2017, **SIAM Conference on Optimization**, Vancouver, British Columbia, Canada (**invited**)
- 12 Apr 2017, **KAUST Research Conference 2017: Visual Computing - Modeling and Reconstruction**, KAUST, Saudi Arabia
- 12 Jan 2017, **SIAM UKIE Annual Meeting 2017**, Glasgow, UK
- 09 Dec 2016, **IEEE Global Conference on Signal and Information Processing (GlobalSIP)**, Washington D.C, USA
- 12 Oct 2016, **Data Science Research Day 2016**, Edinburgh, UK
- 07 Sept 2016, **5th IMA Conference on Numerical Linear Algebra and Optimisation**, Birmingham, UK, (**organizer of 2 mini-symposia**)
- 26 May 2016, **3DT and Friends Student Conference**, Edinburgh, UK
- 16 May 2016, **Machine Learning Summer School (MLSS)**, Cádiz, Spain
- 23 Feb 2016: **5th International Conference on Operational Research and Enterprise Systems (ICORES)**, Rome, Italy